

Payment Processing

International Card Market - Opportunity Revisited; Growth Potential Outweighs Risks

2025 - Extel Institutional Investor
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Visa and Mastercard are increasingly pursuing international opportunities to drive total company bankcard volume growth (non-U.S. volume contributed $\frac{2}{3}$ of Visa's FXN volume growth last year, $\frac{3}{4}$ for Mastercard) as the United States market matures and domestic cash-to-card conversion tailwinds ease. The international market comes with its own set of hurdles, including nationalism and domestic scheme competition, as well as opportunities such as bankcard penetration of PCE in under-penetrated developed and emerging markets. We remain bullish on the networks' value propositions and right-to-win globally, despite incremental geopolitical turmoil that could pressure local banks and regulators to move more swiftly towards network independence. We see \$4-5T incremental V/MA volume opportunity from directly addressable international markets, supplemental to underlying consumption growth in these regions which is nearly 2x domestic growth rates.

- **Global growth breakdown.** Network U.S. bankcard volume growth of 6% y/y lagged non-U.S. growth of 12% in 2024, consistent with expectations and history as more-mature markets' card conversion tailwinds ease. LatAm continued its strong relative volume growth at +22% y/y (or a +16% premium to U.S. growth vs +12% 18-yr median). This report highlights the Brazilian card market (LatAm Financials Team, Yuri Fernandes), which we view as an attractive growth area for the networks and others. European volume growth performed relatively well in 2024 at +15% y/y (+9% premium to U.S. vs +4% median), while Canada, APAC, and the U.S. have all seen volume growth decelerate since 2022 (to mid- to high-single-digits). The U.S. remains by far the largest market for V/MA card volume at \$9T vs. the next-largest region, Europe, at \$5T and next-largest country, the UK, at ~\$1T, by our analysis.
- **Incremental international opportunity.** We examined the top 20 countries by GDP globally to identify the under-penetrated developed markets that present the greatest runway for V/MA intl. volume growth (developed markets defined as \$30k+ GDP per capita and under-penetrated markets defined as V/MA volume % of GDP <20%). The eight under-penetrated developed markets that we identified could present nearly a \$2.0T incremental purchase volume opportunity for V/MA, or nearly a 10% addition to 2024 global V/MA purchase volume. Emerging markets, while more difficult to penetrate, could present another \$2.6T (assuming 15% penetration vs. 20% for developed markets). As a reminder, this volume TAM is incremental to the underlying consumption growth in these economies, which is running almost 2x the domestic growth rate.
- **Nationalism and domestic schemes.** We discuss risks to intl. volume growth

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within, specifically nationalism, or the desire for local actors to reduce dependence on U.S. networks by overseeing their own payment schemes. Nationalism and domestic schemes are not new; local payment architectures like China's UnionPay and Canada's Interac have operated for decades, as have bank-based payment alternatives like Swish in Sweden and iDEAL in the Netherlands. We examine recent trends for these alternative systems within. More abundant geopolitical turmoil, including escalating trade policy tensions on tariffs or sanctions, intensifies the motivation for more independent, domestic control of financial systems, including payment networks, all across the world. That said, Mastercard and Visa have long navigated the ebbs and flows of geopolitics, and we don't anticipate a structural threat to their global expansion.

International Card Market Overview

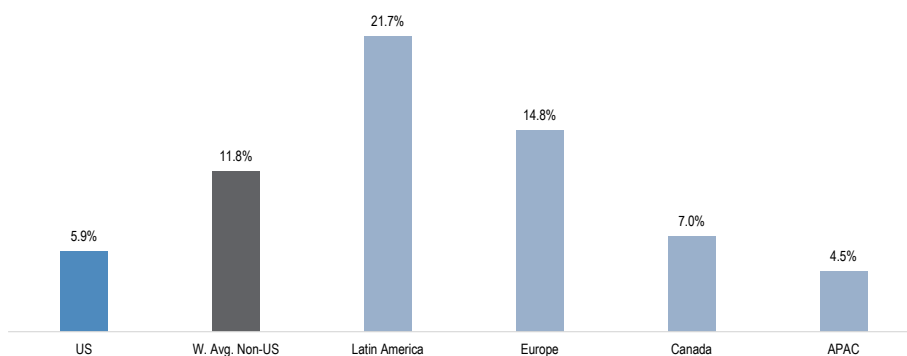
In 2024, U.S. V/MA volume grew 6%, while the rest of the world grew volume twice as fast at 12%, driven by strong performance from LatAm and Europe. We expect non-U.S. volume to continue to grow at a healthy premium as card penetration tailwinds ease in the U.S. Nationalism, or the desire for local banks and regulators to reduce dependence on U.S. networks by overseeing their own payment schemes, remains one of the biggest risks to the industry's addressable market in our view, though we think the tide of increasingly globalized commerce and the value of interoperable payment methods outweighs this risk.

U.S. Growth Lags International

The figure below shows aggregate Visa/Mastercard constant currency purchase volume growth rates in 2024 by region. Purchase card volume growth was strongest in Latin America and Europe, while Canada volume growth barely edged out U.S. volume growth, and APAC lagged the U.S.

Figure 1: Bank Card Regional Purchase Volume Growth

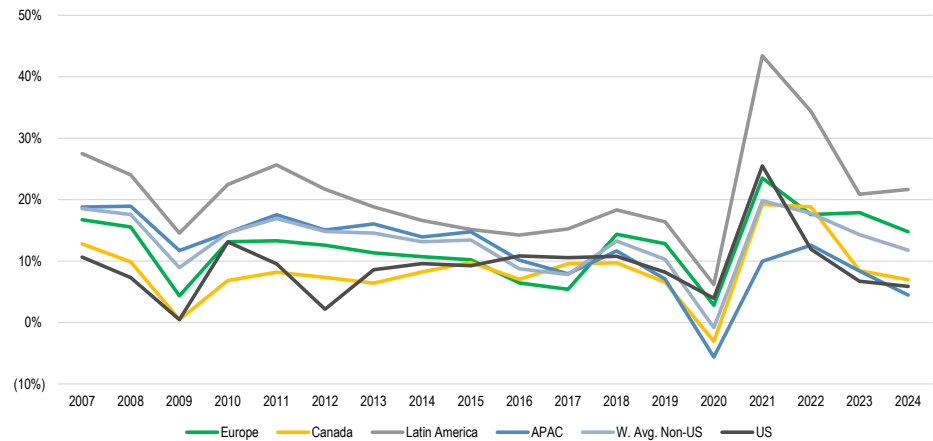
Data for 2024.



Source: Company reports.

LatAm serves as a long-standing driver of non-U.S. volume growth for V/MA. APAC used to be a slight driver of growth for the non-U.S. category, but it has become a laggard since 2018.

Figure 2: Latin America Drives Global Growth; APAC Lags vs A Decade Ago



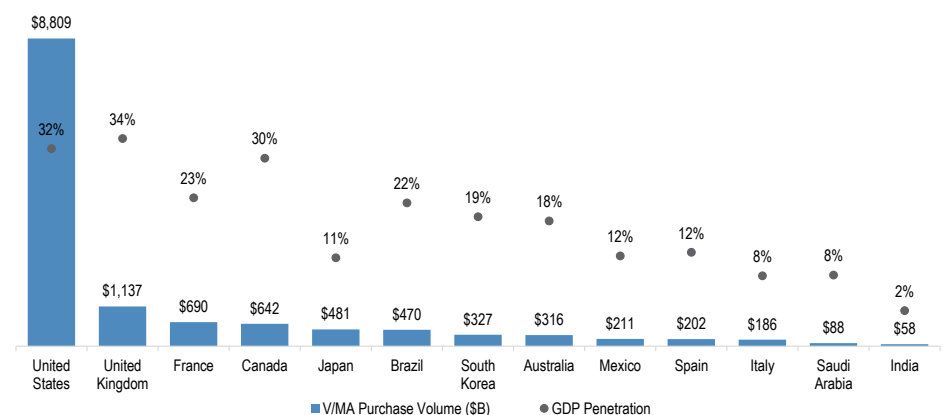
Source: Company reports, J.P. Morgan estimates.

Bank Card Penetration Rates by Region

The figure below shows V/MA purchase volume as a percentage of GDP for key countries. We found that bank card purchase volume represents roughly - 35% of GDP in mature markets like the U.S., U.K., France and Canada, but only 9% -20% of GDP in other markets like Japan, Brazil, South Korea, and Spain. Brazil's volume as a percentage of GDP has experienced the most significant growth in the group, jumping from 4% in 2008 to 22% in 2023.

Figure 3: Bank Card Purchase Volume Penetration Rates

Data for 2023.



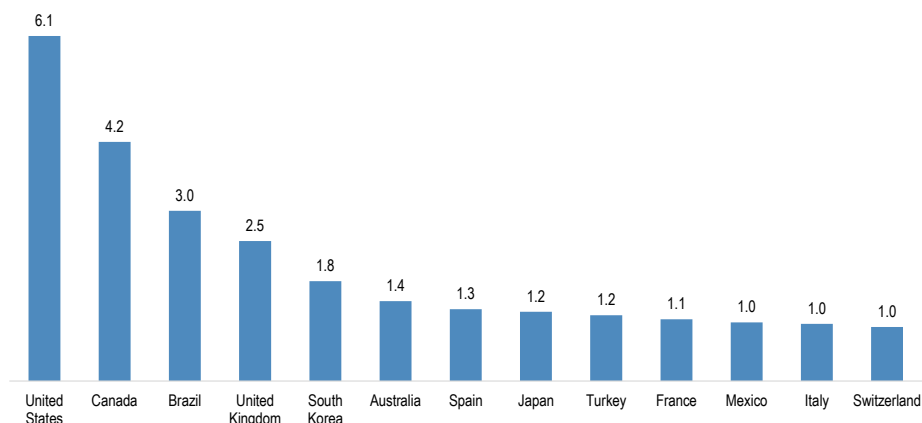
Note: Data excludes Russia and Indonesia. Country-specific bankcard volume (excluding U.S. and Canada) estimated based on data from the regions' top-50 issuers.

Source: Company reports, The Nilson Report, The World Bank, J.P. Morgan estimates.

The figure below shows the bank card per capita ratio across various regions and paints a similar picture. Again, the emerging markets are far less penetrated than the United States, suggesting a long runway for growth ahead. Even among the developed markets, Europe clearly lags North America.

Figure 4: Bank Cards per Capita

Data for 2023.



Note: Data excludes Russia and Indonesia. Country-specific bank cards (excluding U.S. and Canada) estimated based on data from the regions' top-50 issuers, which likely undercounts actual total bank cards.

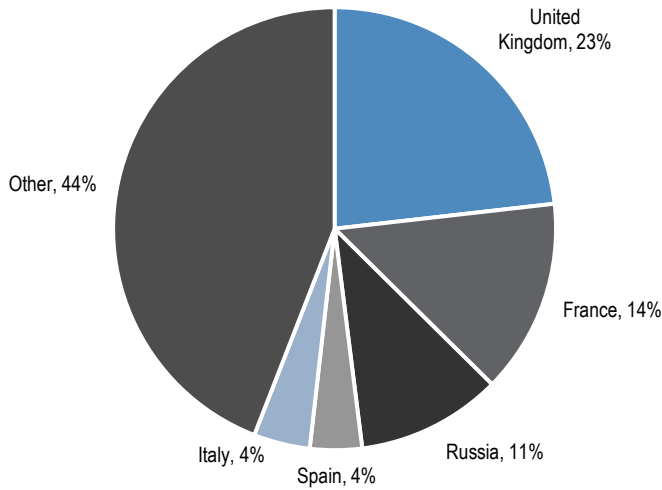
Source: Company reports, The Nilson Report, The World Bank, J.P. Morgan estimates.

A Handful of Key Countries Drive International Volume

Based on our analysis, the bulk of bank card payment volume, particularly in emerging markets, is concentrated in a few countries. Within Europe, the United Kingdom, France and Russia account for around half of the region's bankcard purchase volume (2023 data). In the Asia Pacific region, China alone represents 49% of all bankcard purchase volume, and Japan, South Korea, Australia, and India contribute another ~10%. In Latin America, Brazil and Mexico account for ~54% of bankcard purchase volume. The Middle East and Africa is the most geographically balanced region we examine: the top five countries in the region (Israel, Saudi Arabia, the U.A.E., Kuwait, and Nigeria) account for ~37% of all bankcard purchase volume, and those countries in the region beyond the top five account for the greatest percentage vs the other regions at 63%.

Figure 5: European Bank Card Purchase Volume

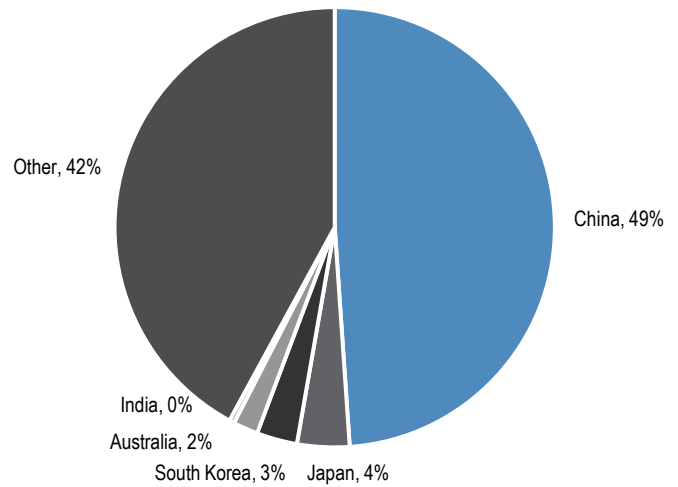
Data for 2023.



Note: Estimated from region's top-50 issuers.
 Source: The Nilson Report, J.P. Morgan estimates.

Figure 6: Asia Pac Bank Card Purchase Volume

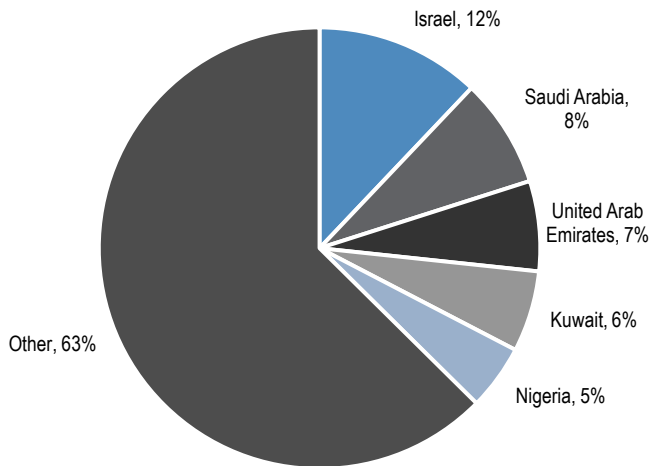
Data for 2023.



Note: Estimated from region's top-50 issuers.
 Source: The Nilson Report, J.P. Morgan estimates.

Figure 7: ME Africa Bank Card Purchase Volume

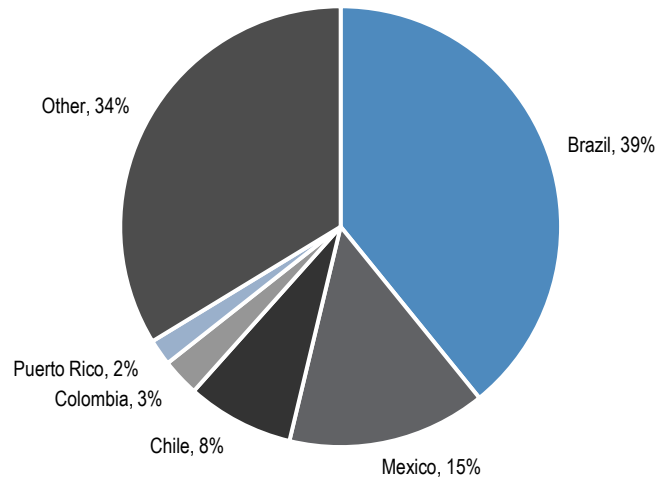
Data for 2023.



Note: Estimated from region's top-50 issuers.
 Source: The Nilson Report, J.P. Morgan estimates.

Figure 8: LatAm Bank Card Purchase Volume

Data for 2023.



Note: Estimated from region's top-50 issuers.
 Source: The Nilson Report, J.P. Morgan estimates.

Snapshot of Domestic Schemes

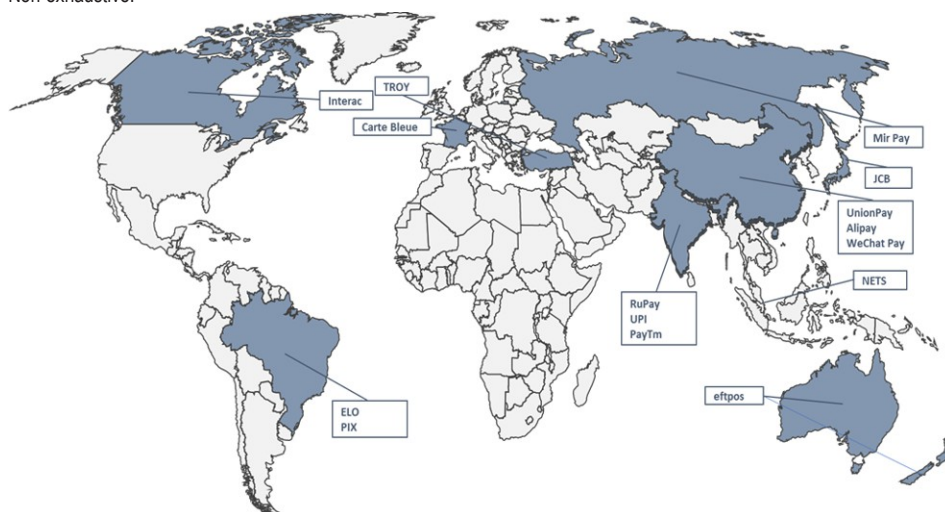
We have long believed that the emergence of domestic schemes overseas is a key long-term threat to the networks, merchant acquirers and payment service providers, fueled by the desire for local banks and regulators to control governance. Economics could be a motivation as well given the large market capitalizations assigned to public payments firms like Mastercard and Visa.

Examples of local schemes include China's UnionPay, Canada's Interac, Russia's Mir by NSPC, India's RuPay, Puerto Rico's ATH Network (owned by Evertec), Australia's eftpos, Turkey's TROY. There are also products like Swish (bank-based mobile payments) in Sweden and iDEAL in the Netherlands that have gained traction as bank-based payment alternatives. We expect interest in developing and promoting local schemes to continue to rise.

The threat of nationalism shouldn't impact our three-year V/MA model, but it could reduce the global TAM (Visa currently excludes domestic China and Russia in its TAM) should local schemes become successful in gaining wallet share in local markets.

Figure 9: Leading Domestic Schemes Globally

Non-exhaustive.



Source: Company reports.

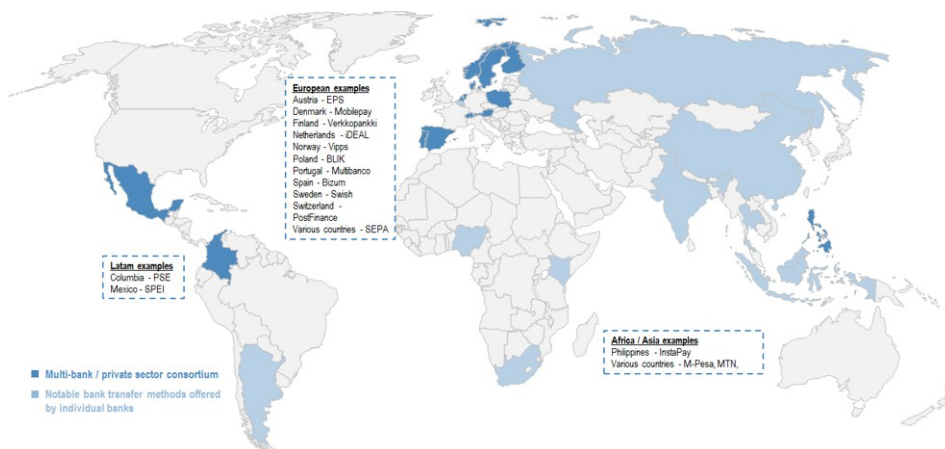
Figure 10: Other Domestic Cards Scheme Examples

Name	Country	Notes
Bancontact	Belgium	17M cards in Belgium; co-branded with Maestro for intl. 2.3B transactions in 2022. 2018 merger with Payconiq
BankAxept	Norway	57% of card transactions in Norway on BankAxept. Enabled for online use in 2017; co-branded with VIMA for intl purchases
BC Card	South Korea	Co-brand with VIMA, JCB and Unionpay for intl use
Carnet	Mexico	Mostly issued by smaller banks with slight skew towards lower income customers
Dankort	Denmark	Debit card scheme that is free for users; co-brand with V for intl purchases. Almost 60% of transactions in Denmark on Dankort card
DinaCard	Serbia	Partnered with Discover and UnionPay; 5M+ cards across 20 issuing banks
Girocard	Germany	Domestic interbank network and debit card system with 100M+ cards; co-brand with VIMA for intl use
Mercury	UAE	Acquired by private entity from NETW LN and Abu Dhabi Bank in 2021; Partnership with Discover for intl purchases
Multicaixa	Angola	Backed by EMIS, which launched a multibank app to support mobile commerce use cases
NAPAS	Vietnam	Domestic debit, credit and prepaid card scheme. Also facilitates QR payment system. Acceptance agreement with multiple APAC networks
PagoBancomat	Italy	Introduced in 1983 as an ATM network; co-branded with VIMA for intl purchases. 32M cards as of 2023
PayPak	Pakistan	Launched in 2016; debit scheme with 5M+ cards issued as of 2021 (~10% of market)
PostFinance	Switzerland	Also operates a more prevalent pay-by-bank option; Card scheme co-branded with Mastercard
Verve	Nigeria	Co-brand or partnerships with VIMA/Discover; available in 4+ countries

Source: JPM Payments Insights, company reports, J.P. Morgan research.

Note: non-exhaustive.

Figure 11: Notable Bank-Transfer-Based Alternative Payment Schemes



Source: Company reports, J.P. Morgan research.

Note: Above is a non-exhaustive list of notable examples; List of banks providing e-commerce checkout flows include: Russia (Sberbank), Belgium (Belfius), Kenya and others (M-Pesa / Safaricom), South Africa (FNB, Nedbank, Standard Bank, Capitec, etc.), China (Bank of China, Agricultural Bank of China, etc.), Indonesia (BCA, Bank Bri, Mandiri, BNI, Maybank, etc.), Malaysia (Maybank, CIMB, PBeBank, etc.), Nigeria (Ecobank, Access, Sterling Bank, etc.).

China is Opening but Remains Elusive to Foreign Firms

China is a \$15+ trillion payment market that has been largely off limits to foreign payment networks, affording UnionPay a monopoly on domestic bankcard payment processing. In May 2015, China first announced it would allow foreign firms to switch and clear domestic bankcard volume, but regulatory hurdles have delayed foreign entry. In June 2020, the PBOC granted American Express's China joint venture (Express Technology Services Company, a JV with LianLian DigiTech) a clearing license for domestic transactions. Meanwhile, PayPal acquired the remaining stake in GoPay in late 2019, giving it 100% control of the online payment platform in China.

The China opportunity is large, but it is unclear just how real the opportunity is for Mastercard and Visa, and whether either company will be able to unseat incumbents (e.g., UnionPay, Alipay and Tencent) and earn an adequate return on investment while satisfying regulatory requirements. We expect progress to be slow. For instance, Mastercard first entered the Chinese market almost 40 years ago and was only granted approval to begin domestic bankcard clearing in 2020. In May 2024, Mastercard's joint venture with NetsUnion processed its first domestic payment in China.

Card fees are lower in China than in the U.S., with merchant interchange fees capped at 0.35% for debit and 0.45% for credit cards. Additionally, competition for high-margin outbound cross-border volume is intensifying as UnionPay's global acceptance grows and co-branded cards are phased out, while the gradual roll-out of the PBOC's digital yuan (via integrations with WeChat and Alibaba) likely reduces the longer-term card penetration opportunity for domestic Chinese payment use cases. As such, we are cautious and continue to monitor the China opportunity.

UnionPay Dominated Domestically...

UnionPay enjoyed a near monopoly on in-country (i.e., domestic) bankcard issuing and payment processing; however, Chinese citizens were allowed to use foreign-branded or co-branded payment cards bearing foreign network logos when transacting abroad. These co-branded cards are being phased out, meaning Chinese citizens now have to use

a separate card when transacting abroad. Within China, tourists can use foreign payment cards (like Mastercard and Visa) wherever they are accepted. Below, we summarize UnionPay's published usage statistics.

Figure 12: UnionPay Statistics

Values in billions; volume shown in USD.

	2016	2017	2018	2019	2020	2021	2022	2023
Debit Purchase Volume	\$5,313	\$6,228	\$7,889	\$7,607	\$7,328	\$8,940	\$8,076	\$8,184
Credit Purchase Volume	\$3,190	\$3,936	\$6,996	\$6,745	\$7,216	\$8,372	\$7,767	\$6,919
Total Volume	\$8,503	\$10,164	\$14,885	\$14,352	\$14,544	\$17,312	\$15,843	\$15,103
% change y/y	NA	20%	46%	-4%	1%	19%	-8%	-5%
Transactions	38.3	58.6	100.6	132.8	152.3	198.1	212.7	228.0
% change y/y	NA	53%	72%	32%	15%	30%	7%	7%
Cards	6.1	6.7	7.6	8.4	9.0	9.2	9.4	9.7
% change y/y	NA	10%	13%	11%	7%	2%	2%	3%

Source: Company reports, The Nilson Report (July 2024).

...Now AliPay and Tenpay Dominate Mobile Payments

AliPay and WeChatPay have emerged as the mobile way to pay individuals and merchants, with 1.3B and 1.1B active users, respectively. Acceptance is broad, leveraging simple QR-code technology that users present to merchants to pay. Adyen partnered with Alipay in 2019 to facilitate payments outside of mainland China for AliExpress, Taobao, Tmall and Alibaba.com, which has helped improve approval rates while shortening settlement cycles.

Select Domestic Market Case Studies

SWISH: Sweden's Integrated Payment System

SWISH is a mobile payment system formed in 2012 by a consortium of six large banks in Sweden, with the Central Bank of Sweden overseeing the system. The service works through a smartphone app, primarily used for P2P transactions, through which the user's phone number is connected to their Swedish bank account. SWISH became popular as a replacement for small-ticket size cash payments (for retail transactions), with a nearly costless transaction compared with alternate solutions like PayPal. SWISH has over 8M total users in Sweden, which represents ~80% of the total population (up from 10% in 2014).

In July 2014, SWISH expanded into retail payments, charging SEK 1.5-2 per transaction. SWISH has also expanded its presence in the e-commerce space by enabling merchants to accept payments for online and offline purchases. Users can also pay merchants by scanning QR codes available at partner merchant stores with the Swish app.

During 2024 SWISH processed ~1.1B transactions, up from 618M in 2020. Swish utilization has climbed from 52% of Swedes in 2016 to ~80%, with over 90% of users sending or receiving a payment every month.

iDEAL: A Low-Cost Network in the Netherlands

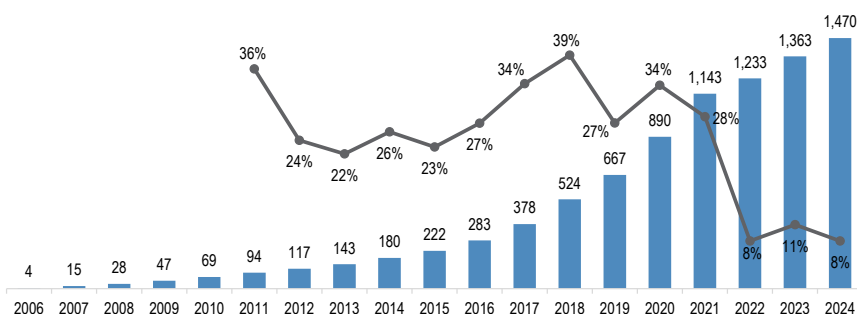
Launched in 2005, iDEAL in the Netherlands is an online mobile payment system offered by a network of 14 issuers, 18 merchant acquirers and almost 60 payment institutions. The platform bypasses traditional card networks, pulling funds directly from users' bank accounts. Customers are automatically enrolled in the program if they bank at one of the 14 participating issuers and as of 2024 can create accounts directly with iDEAL.

iDEAL enjoys a ~70% share of online transactions in the Netherlands (up from 57% in 2019), with most transactions occurring on mobile (~90% of iDEAL payments). The platform enables payments for international transactions with merchants across 60+ countries with conversion rates comparable to other payment solutions. In 2021, roughly ~10% of transactions were cross-border, and it counts nearly 100% of Dutch citizens as users. In April 2024, iDEAL announced iDEAL in3, which lets users spread payments over three transactions.

In October 2023, iDEAL was acquired by the European Payments Initiative (EPI), which aims to implement a standardized uniform payments solution across Europe. EPI has 16 bank members and is backed by the European Central Bank. iDEAL says it will gradually migrate to [Wero](#), a pan-European payment solution launched in mid-2024, with plans to be fully implemented by 2028.

Figure 13: iDEAL Transactions

Millions of transactions and year-over-year growth.



Source: iDEAL statistics.

Russian National System of Payment Cards

In the wake of Crimea-related financial sanctions that disrupted certain Mastercard/Visa payment functions, Russian President Vladimir Putin pushed to establish a national card payment system. In Dec 2015 seven Russian banks came together to pilot the card program, called Mir, with the cards issued by the Russian Central Bank and its subsidiary the National Payment Card System (NSPK). The cards were co-branded with Mastercard, JCB and American Express for circulation in foreign markets.

The Mir cards can facilitate cash deposits and withdrawals, goods and services payments, and card-to-card money transfers. NSPK also began rolling out support for NFC payments for Android 6.0 and above. As of February 2022, there were ~122M Mir cards in circulation, which made up over 25% of Russia's bank card market.

Visa and Mastercard originally signed agreements with the Bank of Russia and NSPC to transfer their local transaction processing to the NSPC and define settlement rules, where they would generate significantly less revenues from domestic Russian purchase volumes given their reduced role, although the cross-border opportunity remained largely intact. However, due to Treasury Department sanctions enacted in March 2022 in the wake of the Russian invasion of Ukraine, Mastercard and Visa announced that they would cease their domestic Russian operations and stop supporting all co-branded cards in any international or cross-border transaction. Per company filings, Russia made up a mid-single-digit percent of 2021 revenues for V/MA, with a majority of this exposure likely coming from cross-border spend where V/MA had earned a higher net

take-rate. In the months following these sanctions, many Russian banks such as Sberbank have studied the possibility of issuing co-branded cards with Unionpay to circumvent sanctions and enable use for international transactions. Given the current uncertain state of US-Russian relations and low visibility on the potential duration of the Ukraine conflict, it is plausible that the Russian market may be closed off to V/MA for the foreseeable future as the NPSK and individual Russian banks explore deeper relationships with China's Unionpay.

ELO and Industry Verticalization in Brazil

The ELO network was created in 2011 as a JV of three of the largest Brazilian Banks: Banco do Brazil (~16% share of loans), Caixa Econômica (~20% share of loans) and Bradesco (~10% share of loans). J.P. Morgan's Latin American Financials analyst Yuri Fernandes estimates all brand revenues in Brazil amounted to \$3.1B-\$3.8B in 2024, from which Mastercard and Visa likely retain ~70%-90% on a gross basis. For details on its strategy please see: [“Lessons from Elo – TPV Outlook, PIX inside Cards, Meal Voucher More.”](#)

Currently, we estimate the ELO network has ~9% market share of volumes, mostly concentrated in debit cards (~23% market share). In past years Elo has been losing market share – the company had over 80mn active debit cards in 2020, or 42% market share, helped by government social transfers during Covid-19. Recently, the market share of active debit cards decreased to ~23% in 1H24, or some 37.4mn active debit cards, about half the share it had a few years ago. We believe the market share lost is explained by the end of those programs combined with a growing market share of neobanks like Nubank, which issue their debit / prepaid cards using Mastercard or Visa vs. a weaker franchise of controlling banks, which were, and still are, the main distributors of ELO cards.

On credit cards, the company has 9.2mn active plastics, or ~4% of total, also down from ~6% share in 2020. In credit, the main strategy for growth is the offering of more compelling loyalty programs, which yield more points per US\$ spent vs. Visa and Mastercard cards of the same category. Additionally, ELO has an agreement with Discovery that allows ELO cards being accepted outside of Brazil.

Until mid-2016, Cielo was the only merchant acquirer able to capture ELO transactions. After 2016, other acquirers started being able to capture those transactions but routing them to Cielo, earning a flat fee per transaction. Today, most acquirers act as full acquirer model and thus earning net MDRs. This transition boosted ELO's acceptance network, thus leading to stronger growth.

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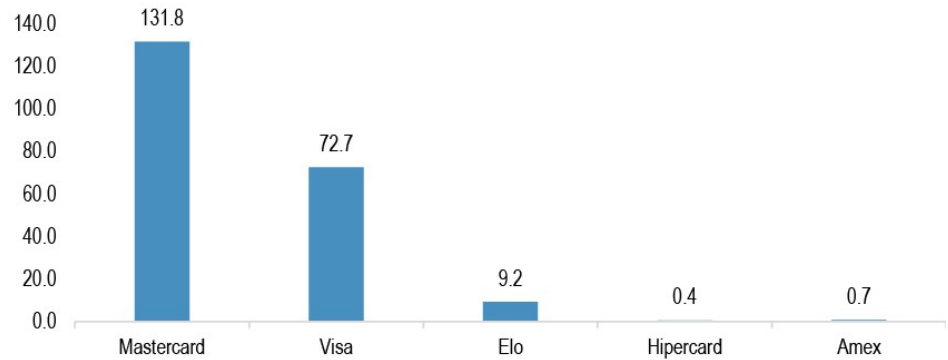
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Figure 14: Active Brazilian Credit Cards

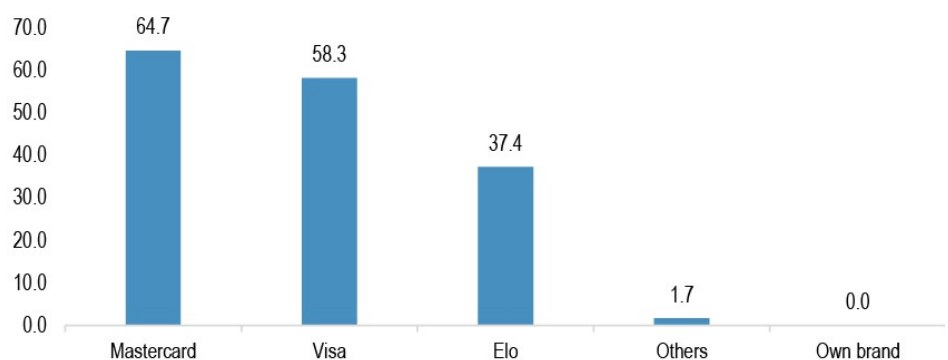
Data as of 1H24.



Source: Brazil Central Bank.

Figure 15: Active Brazilian Debit Cards

Data as of 1H24.



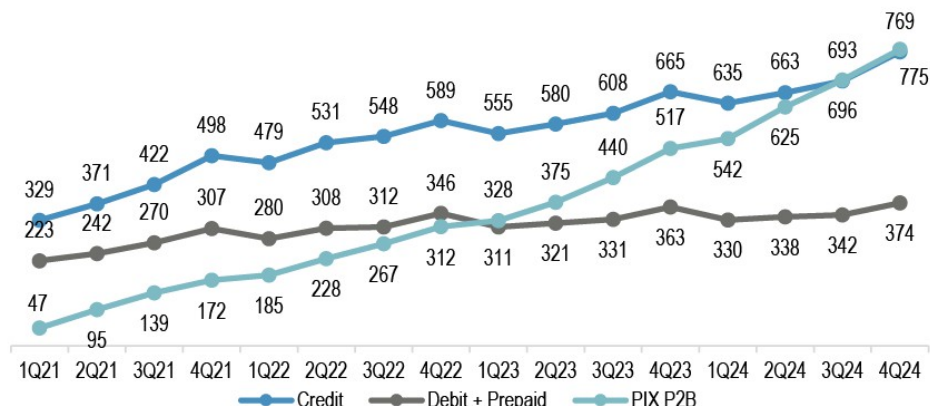
Source: Brazil Central Bank.

Brazil's Instant Payments PIX

In November 2020, Brazilian Central Bank officially launched its instant payments platform PIX, a 24/7 payment platform free of charge for P2P. Since then, the government's rail has been gaining rapid share on P2P transactions, and more recently also on P2B payments. Notably, today the PIX platform has 158mn registered SSN (social security numbers), or ~75% of Brazil's population; and ~16mn companies. For the last update on it, please see: ["PIX 4 year update – 10 takes an New Functionalities overview."](#)

Total notional value transacted has also been growing rapidly, with ~R\$26 trillion in volume transacted in 2024 vs ~R\$17 trillion in 2023. Most of the market share gains have been over bank ACH and Wire, but some cannibalization of debit volumes has taken place too. Specifically, PIX volumes for P2B already surpassed the size of debit + pre-paid card volumes in Brazil (R\$2.6tn in 2024 vs R\$1.4tn debit + prepaid). Additionally, we note select long-tail merchants using PIX bank transfers (P2P) for the sale of products and services (e.g., to avoid MDR fees, taxes) hence the disclosed P2B TPV likely underestimates PIX real penetration on merchants. Our estimates point that PIX P2B in practice could be ~2-3x larger than reported ([link](#)).

Figure 16: PIX TPV vs Credit Cards and Debit Cards (incl. Pre-Paid Cards)



Source: Brazil Central Bank, ABECS.

Interac: Canada's Domestic Debit Network

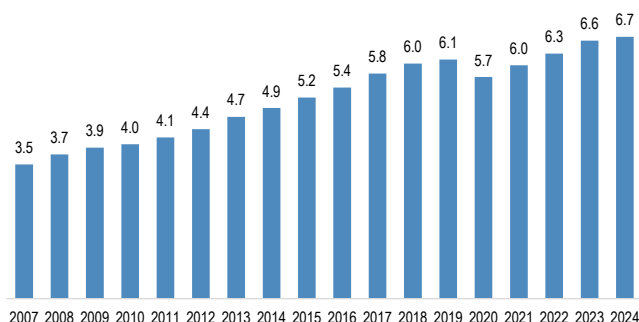
Interac is the most widely accepted debit PIN network in Canada, while V/MA volume still primarily represents credit. Interac offers a shared network for cash dispersion from automated banking machines (ABMs) and a shared network to allow debit cardholders to pay for purchases at POS. Interac Debit also includes Interac Flash, a secure contactless enhancement that allows Canadians to pay for items instantly by holding their cards at point-of-sale readers that support Interac Flash. Additionally, Interac offers an integrated P2P service, letting users send/request funds from account to account using a phone number or email address.

Visa and Mastercard have been aggressively trying to grow their debit presence in Canada and have been making progress since 2013. In terms of purchase transaction market share, Visa now comprises 37% of the Canadian market, while Mastercard comprises 24% (as per Nilson report).

The figures below show historical transactions and volume on Canada's Interac network. Interac transactions and card volume have both grown at 5% CAGRs since 2003 (or 3% and 4%, respectively, in the last 10 years).

Figure 17: Interac Transactions

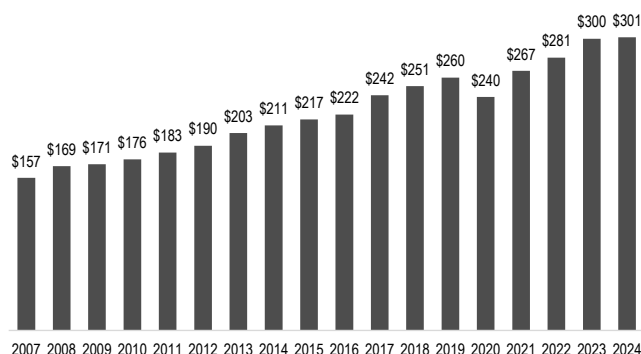
Transactions in billions.



Source: Company reports, The Nilson Report.

Figure 18: Interac Card Volume

CAD in billions.



Source: Company reports, The Nilson Report.

Sizing the International Opportunity

The Largest and Most Developed Economies

In this section we largely leverage 2023 data, as some sources are released on a one year lag.

The table below shows economic, consumption, and payment penetration data for the 20 largest economies in the world by GDP. These countries generated 79% of global GDP in 2023 and represented 54% of the world's population.

Within this subset, we leveraged a rough bar of ~\$30k annual per capita GDP to identify the most developed nations. The 12 “developed” countries, shaded in gray, generated 51% of aggregate global GDP in 2023 (up 1ppt from 2022) while only representing 12% of the world's population (stable y/y). We believe these “developed” countries represent the most tangible and immediate cash conversion opportunity for Mastercard and Visa due to their powerful consumers, elevated cards per capita, and high cardable GDP opportunities.

Figure 19: Largest Economies by GDP

Data for 2023.

Country	GDP (billions USD)	Population (millions)	Household Consumption (% GDP)	GDP per Capita	V/M Vol. (billions)	Cards (millions)	Cards per Capita	Payment Penetration	Payment Penetration (Consumption)
United States	\$27,721	335	68%	\$82,769	\$8,809	2,050	6.1	32%	47%
China	\$17,795	1,411	39%	\$12,614	\$710	267	0.2	4%	10%
Germany	\$4,526	83	50%	\$54,343	\$73	17	0.2	2%	3%
Japan	\$4,204	125	56%	\$33,767	\$481	153	1.2	11%	20%
India	\$3,568	1,438	60%	\$2,481	\$58	207	0.1	2%	3%
United Kingdom	\$3,381	68	61%	\$49,464	\$1,137	170	2.5	34%	55%
France	\$3,052	68	53%	\$44,691	\$690	75	1.1	23%	42%
Italy	\$2,301	59	58%	\$39,003	\$186	60	1.0	8%	14%
Brazil	\$2,174	211	63%	\$10,295	\$470	638	3.0	22%	34%
Canada	\$2,142	40	55%	\$53,431	\$642	170	4.2	30%	54%
Mexico	\$1,789	130	70%	\$13,790	\$211	135	1.0	12%	17%
Australia	\$1,728	27	50%	\$64,821	\$316	38	1.4	18%	36%
South Korea	\$1,713	52	49%	\$33,121	\$327	92	1.8	19%	39%
Spain	\$1,620	48	54%	\$33,509	\$202	62	1.3	12%	23%
Netherlands	\$1,154	18	42%	\$64,572	\$25	10	0.5	2%	5%
Turkey	\$1,118	85	59%	\$13,106	\$128	100	1.2	11%	19%
Saudi Arabia	\$1,068	33	40%	\$32,094	\$88	27	0.8	8%	21%
Switzerland	\$885	9	52%	\$99,565	\$59	9	1.0	7%	13%
Poland	\$809	37	58%	\$22,057	\$72	17	0.5	9%	15%
Argentina	\$646	46	66%	\$14,187	\$11	12	0.3	2%	3%
Total Top 20	\$83,393	4,322	56%	\$19,293	\$14,697	4,305	1.0	18%	31%
World	\$106,172	8,062	56%	\$13,170	\$19,938	7,418	0.9	19%	33%
Top 20 Developed	\$54,427	932	61%	\$58,403	\$12,948	2,904	3.1	24%	39%
Top 20 Developing	\$28,966	3,390	47%	\$8,543	\$1,748	1,402	0.4	6%	13%

Note: Data excludes Russia and Indonesia. Payment penetration is defined as estimated V/M volume / GDP. Payment penetration (consumption) is defined as estimated V/M volume / household consumption expenditures. Household consumption is the market value of all goods and services, including durable products, as defined by the World Bank.

Source: The Nilson Report, World DataBank, Visa/Mastercard company reports, J.P. Morgan estimates.

Identifying “Under-Penetrated” Developed Economies

Eight of the eleven “developed” countries (excluding the United States) have payment penetration rates at or below 20% (with penetration defined as V/M volume over GDP). These “under-penetrated” countries (highlighted in gray in the table below) generate ~\$18 trillion in aggregate GDP (65% of the United States) and an estimated ~\$1.7 trillion in annual Visa/Mastercard purchase volume, representing an aggregate payment penetration rate of only 9%. Assuming a 20% penetration rate (a conservative discount to the United States where Visa/Mastercard purchase volume represents ~33% of GDP), **these eight “under-penetrated” countries could represent more than a \$2.0 trillion incremental purchase volume opportunity for Visa and Mastercard**, or ~10% of Mastercard and Visa’s 2023 global purchase volume figure. Said differently, each point of payment penetration of GDP within this group would represent as much as \$200B in incremental volume for Visa and Mastercard.

Figure 20: Payment Penetration in Major Developed Countries

Data for 2023.

Country	GDP (billions USD)	Population (millions)	Household Consumption (% GDP)	GDP per Capita	V/MA Vol. (billions)	Cards (millions)	Cards per Capita	Payment Penetration	Payment Penetration (Consumption)
Germany	\$4,526	83	50%	\$54,343	\$73	17	0.2	2%	3%
Japan	\$4,204	125	56%	\$33,767	\$481	153	1.2	11%	20%
United Kingdom	\$3,381	68	61%	\$49,464	\$1,137	170	2.5	34%	55%
France	\$3,052	68	53%	\$44,691	\$690	75	1.1	23%	42%
Italy	\$2,301	59	58%	\$39,003	\$186	60	1.0	8%	14%
Canada	\$2,142	40	55%	\$53,431	\$642	170	4.2	30%	54%
Australia	\$1,728	27	50%	\$64,821	\$316	38	1.4	18%	36%
South Korea	\$1,713	52	49%	\$33,121	\$327	92	1.8	19%	39%
Spain	\$1,620	48	54%	\$33,509	\$202	62	1.3	12%	23%
Netherlands	\$1,154	18	42%	\$64,572	\$25	10	0.5	2%	5%
Switzerland	\$885	9	52%	\$99,565	\$59	9	1.0	7%	13%
Under-Penetrated	\$18,131	420	52%	\$47,616	\$1,670	439	1.0	9%	17%
United States	\$27,721	335	68%	\$82,769	\$8,809	2,050	6.1	32%	47%

Note: Payment penetration is defined as estimated V/MA volume / GDP. Payment penetration (consumption) is defined as estimated V/MA volume / household consumption expenditures. Country-specific V/MA volume (excluding US and Canada) is estimated based on data from the regions' top 50 issuers, which likely undercounts bankcard volume.

Source: The Nilson Report, World DataBank, Visa/Mastercard company reports, J.P. Morgan estimates.

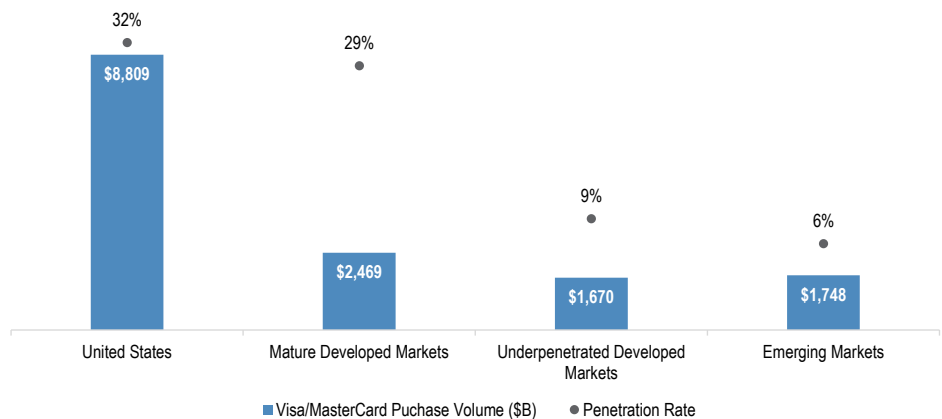
Under-Penetrated and Emerging Markets Could Represent a \$4.6 Trillion Purchase Opportunity

The figure below shows Visa/Mastercard purchase volume and penetration rates classified by penetration and “development” levels. We define developed markets as those with ~\$30k+ GDP per capita; within developed markets, we define under-penetrated developed markets as those with <20% V/MA purchase volume GDP penetration and mature developed markets as those with ≥20% GDP penetration. We define everything else as emerging markets.

We size the “greenfield” purchase volume opportunity in under-penetrated developed and emerging markets at \$2.0 trillion and \$2.6 trillion, respectively.

In 2023, the United States generated \$8.8 trillion in purchase volume through Visa and Mastercard, or 32% of GDP, while other mature developed markets generated \$2.4 trillion in purchase volume, or roughly 29% of their GDP. Both categories have seen a high-single-digit PPT increase in payment penetration since 2007. Under-penetrated developed markets, where we think Mastercard/Visa have the most immediate cash conversion opportunity, generated an estimated \$1.7 trillion in purchase volume, or roughly 9% of their GDP. Finally, emerging markets generated \$1.7 trillion in purchase volume, or roughly 6% of their GDP. We size the “greenfield” purchase volume opportunity in under-penetrated developed and emerging markets at \$2.0 trillion and \$2.6 trillion, respectively, using a 20% and 15% terminal penetration rate, which assumes emerging market penetration does not reach developed levels.

Figure 21: Visa/Mastercard Purchase Volume and Penetration Rates by Penetration Level
Data for 2023.



Note: Data excludes Russia and Indonesia. Country-specific V/MA volume (excluding US and Canada) is estimated based on data from the regions' top 50 issuers, which likely undercounts actual bankcard volume.
Source: The World Bank, The Nilson Report, company reports, J.P. Morgan estimates.

We note that these volume opportunities are incremental to underlying personal consumption growth in these markets, which was up 7% in 2023, nearly 2x the typical growth rate for U.S. consumption.

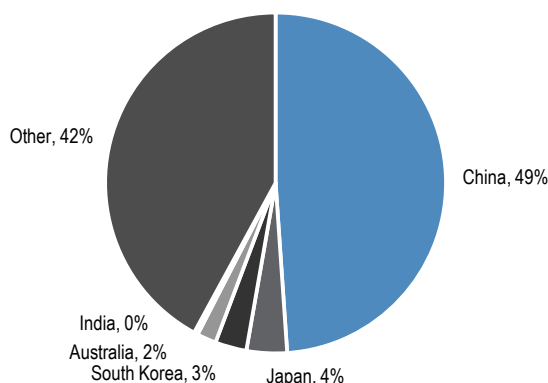
International Payment Markets

The Asia Pacific Payments Market

Asia Pacific generated about \$19 trillion in total purchasing volume in 2023 (including Visa, Mastercard, UnionPay, JCB, Amex and Diners Club), representing just more than 2x the size of the United States market. Visa and Mastercard branded cards made up ~\$3.8 trillion of APAC purchase volume, or ~20% share in 2023. China UnionPay brand cards have a significant market presence in Asia Pacific, with roughly \$15 trillion in APAC purchase volume in 2023, or almost 80% share. We note that UnionPay purchase spend is skewed upward, as it can encompass automobile and house purchase activities, unlike other typical card spend. The figure below estimates market share by country, extrapolating top 50 issuer data from the region. Overall, APAC V/MA volume penetration (over GDP) is only ~10%, the lowest penetration of all the regions examined in this report.

Figure 22: Asia Pacific Purchase Volume by Country

Estimated share for 2023.



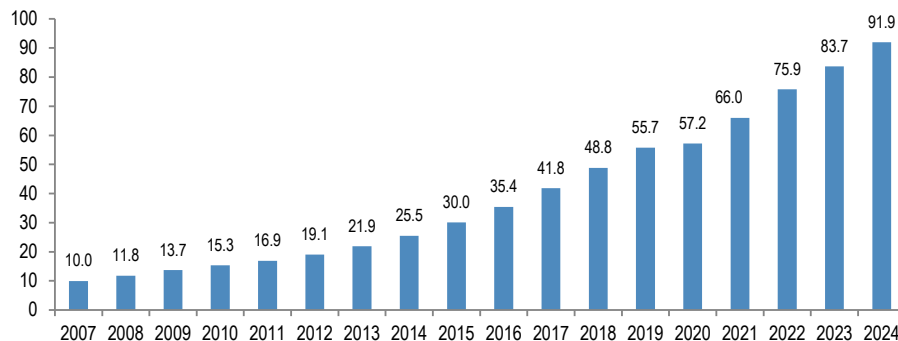
Note: Country-specific bankcard volume estimated based on data from the region's top-50 issuers.

Source: The Nilson Report, company reports, J.P. Morgan calculations.

The figures below show Asia Pacific bankcard (Visa and Mastercard) purchase volume and transaction trends, respectively. We note bankcard purchase volume has increased at a 10% annualized rate since 2007, while bankcard transactions have grown at a 14% annualized rate during the same respective time period.

Figure 23: Asia Pacific Bankcard Transactions

Transactions in billions.



Source: Visa and Mastercard company reports.

Figure 24: Asia Pacific Bankcard Volume

\$ in USD trillions.

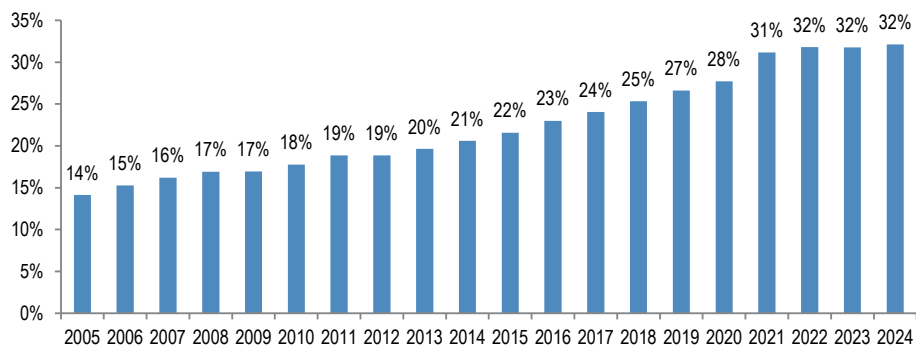


Source: Visa and Mastercard company reports.

The figures below show Asia Pacific payment penetration and cards per capita. Payment penetration is calculated as V/MA purchase volume divided by GDP, while cards per capita is defined as total V/MA cards divided by the population over 15 years old. Payment penetration has grown at a 4% annualized rate since 2005, while cards per capita have grown 7% annually. The V/MA cards per capita metric in Asia Pacific remains the lowest among the different regions we examined, likely due to the majority share of Union Pay cards in the region (over 9B cards vs. less than 2B for Mastercard and Visa combined).

Figure 25: Asia Pacific Payment Penetration

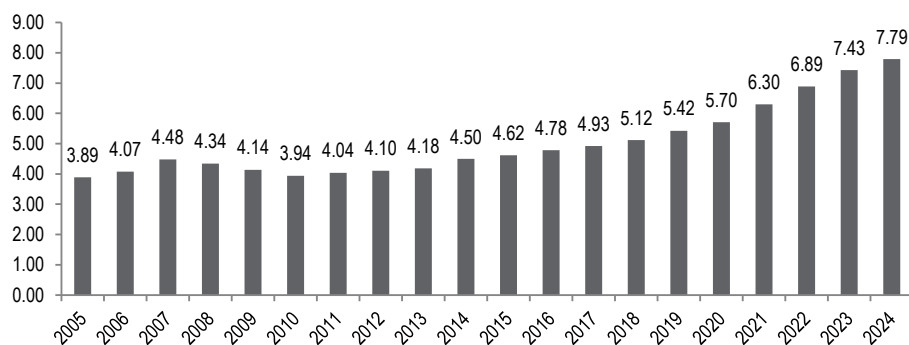
V/MA Volume % of GDP.



Note: For comparison to network disclosure, APAC GDP includes emerging and developing Asia, Australia, Hong Kong SAR, Japan, Korea, Taiwan, Macao SAR, and Singapore.

Source: Company reports, the IMF, J.P. Morgan estimates.

Figure 26: Asia Pacific Cards per Capita

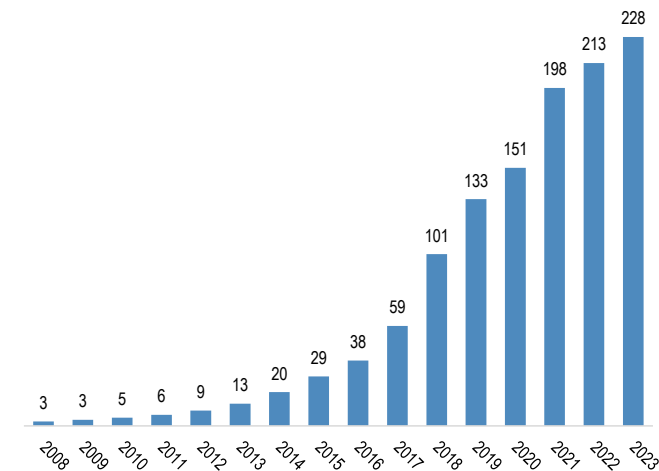


Source: The World DataBank, company reports, J.P. Morgan estimates.

The figures below show transactions and purchase volume on China's UnionPay network. We note that UnionPay purchase volumes have grown at a 27% CAGR since 2008, while transactions have grown at a 37% CAGR. This compares with Visa and Mastercard global volume and transactions growing at 10% and 13% CAGRs, respectively.

Figure 27: UnionPay Purchase Transactions

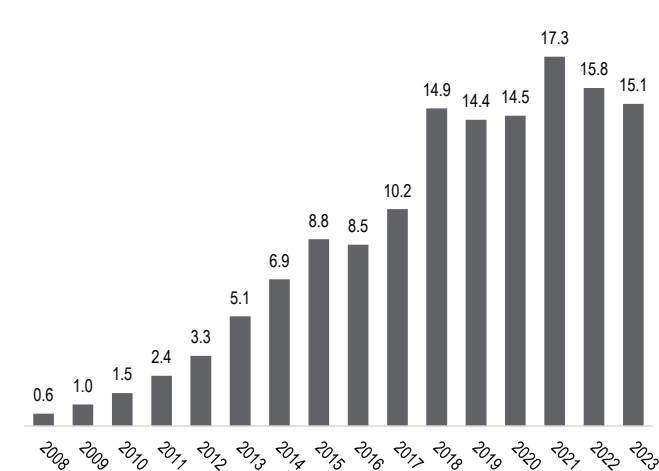
Transactions in billions.



Source: The Nilson Report.

Figure 28: UnionPay Purchase Volume

\$ in USD trillions.



Source: The Nilson Report.

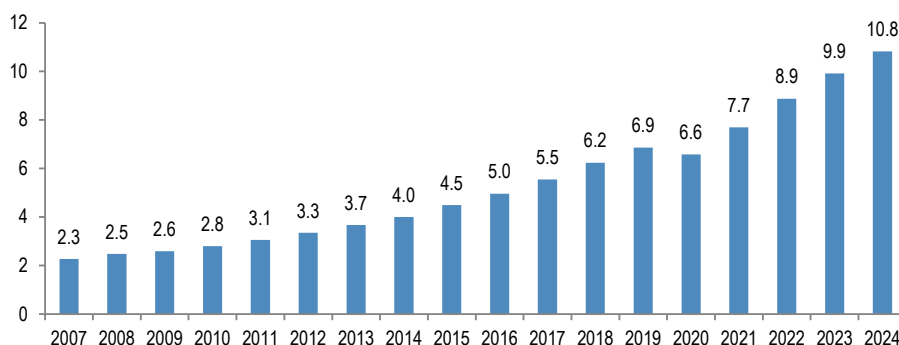
The Canadian Payments Market

Canada generated ~\$900 billion in total purchase volume and 16.8 billion transactions in 2023 (Visa, Mastercard, Amex, Interac), making it about one-tenth the size of the United States market. The Canadian market is unique in that the vast majority of its debit volume is processed over the Interac PIN network (state owned/operated), and Visa and Mastercard volume are still primarily credit. Visa and Mastercard have been aggressively trying to grow their debit presence in Canada and have been making progress since 2013. In terms of total purchase transaction market share, Visa now comprises 37% of the Canadian market, while Mastercard comprises 24%, vs. Interac at 38% (per the Nilson report). Total card-based payment penetration (V/MA, Interac and Amex) is fairly high at 42% of GDP. V/MA volume penetration, shown in the figures below, was 31% 2024, 1ppt below the United States.

The figures below show Canadian V/MA purchase volume and transaction trends. We note bankcard purchase volume has increased at an 8% annualized rate since 2005, while bankcard transactions have grown at a 10% annualized rate during the same respective time period.

Figure 29: Canada Bankcard Transactions

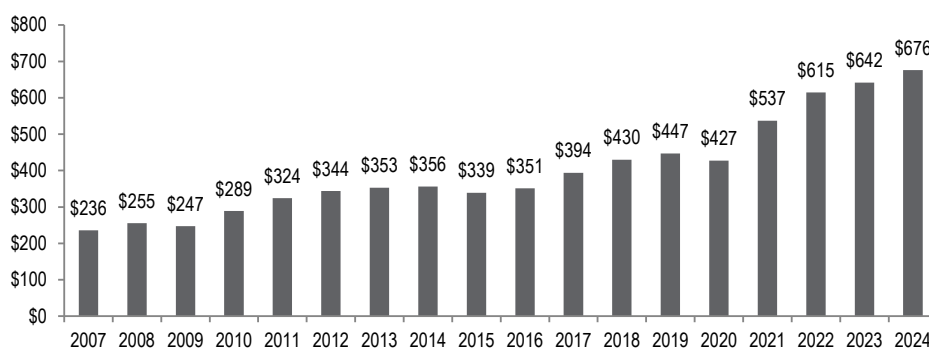
Transactions in billions.



Source: Visa and Mastercard company reports.

Figure 30: Canada Bankcard Purchase Volume

\$ in USD billions.



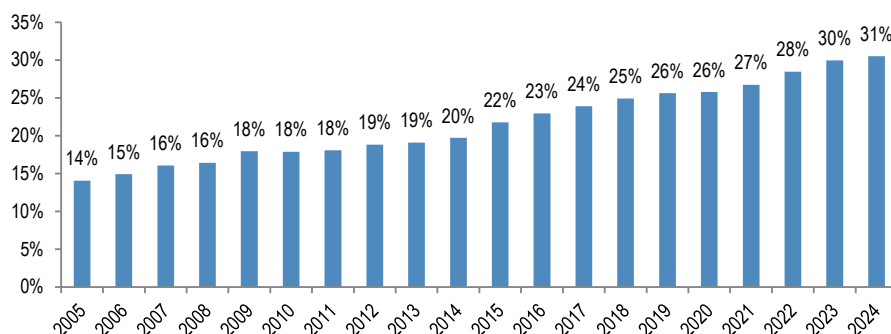
Source: Visa and Mastercard company reports.

The figures below show historical Canadian payment penetration and cards per capita. Payment penetration is calculated as V/MA purchase volume divided by GDP, while cards per capita is defined as total V/MA cards divided by the population over 15 years

old. Payment penetration and cards per capita have grown at 4% and 5% annualized rates since 2005, respectively. Canadian payment penetration has run at a similar rate to that of the U.S. but is still lagging in cards per capita (~5 vs. 7 in U.S.) despite a step-up and acceleration since in 2018.

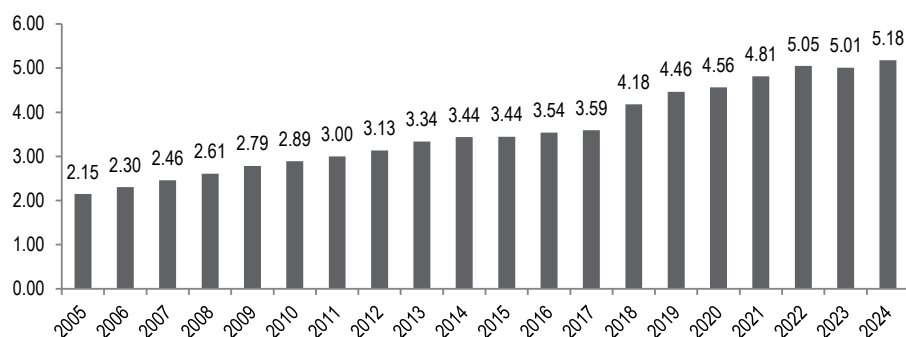
Figure 31: Canada Payment Penetration

V/MA Volume % of GDP.



Source: The IMF, company reports, J.P. Morgan estimates.

Figure 32: Canada Cards per Capita

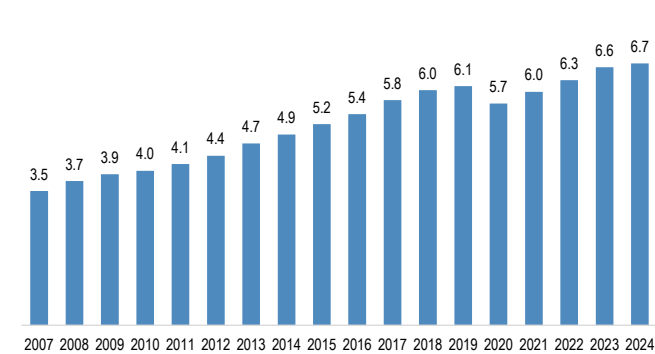


Source: The World Bank, company reports, J.P. Morgan estimates.

The figures below show volume and transactions on Canada's Interac network. We note that Interac transactions and card volume have grown at 5% CAGRs since 2003. According to Interac, Interac Debit on mobile grew 47% in 2023, while Interac e-transfers grew in the teens. According to its 2023 year in review, Interac revenue grew 13%.

Figure 33: Interac Transactions

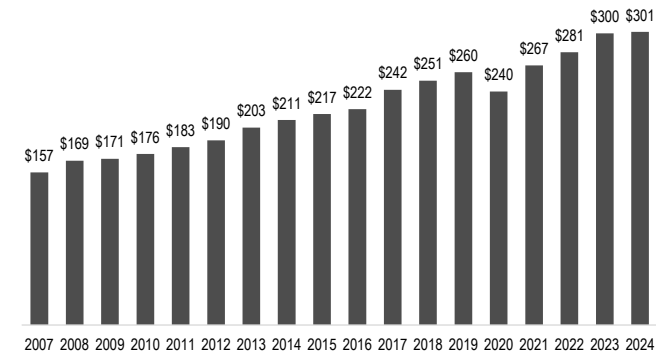
Transactions in billions.



Source: Company reports and The Nilson Report.

Figure 34: Interac Card Volume

\$ in CAD billions.



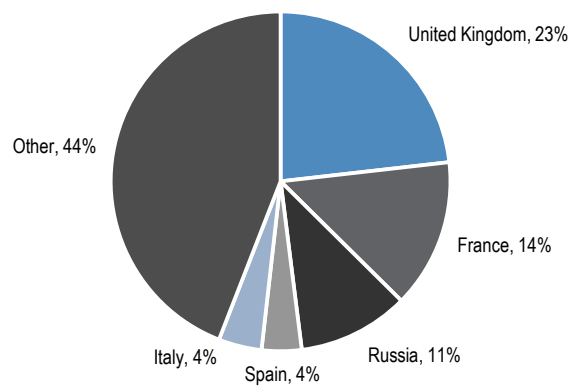
Source: Company reports and The Nilson Report.

The European Payments Market

According to The Nilson Report, Europe generated nearly \$5.0 trillion in purchase volume on 1.6 billion cards in 2023 (including V, MA, Amex and Diners Club). European bankcard payment penetration (V/MA volume over GDP) is approximately 18%, 14ppts below that in the U.S. According to our analysis, U.K. and France are the largest markets for payment volume in Europe, responsible for over a third of all European bankcard volume.

Figure 35: European Purchase Volume by Country

Estimated share for 2023.

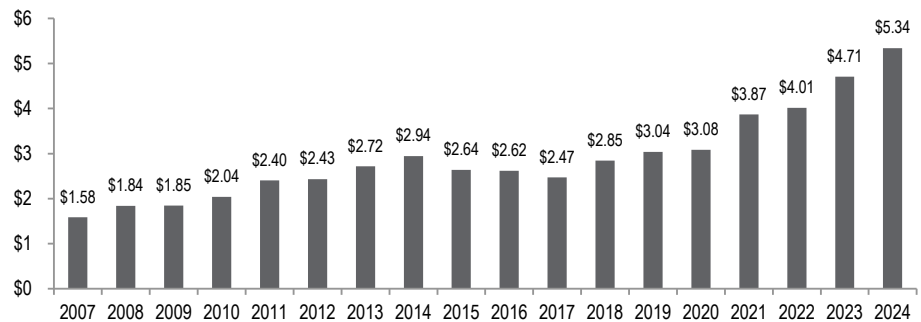


Note: Country-specific bankcard volume estimated based on data from the region's top-50 issuers.
Source: The Nilson Report, company reports, J.P. Morgan calculations.

The figure below shows European bankcard purchase volume. We note purchase volume, when measured in USD, has increased at a 7% annualized rate since 2007.

Figure 36: European Bankcard Purchase Volume

\$ in USD trillions.

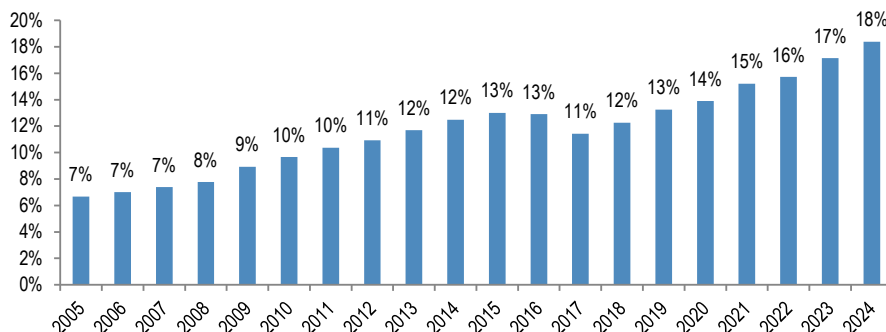


Note: Measured as Visa and Mastercard reported Europe volumes.
Source: Company reports, J.P. Morgan estimates.

The figures below show historical European payment penetration and cards per capita. Payment penetration is calculated as V/MA purchase volume divided by GDP, while cards per capita is defined as total cards divided by the population over 15 years old. Payment penetration has grown at a 5% annualized rate since 2005, two percentage points below card per capita growth of 7%.

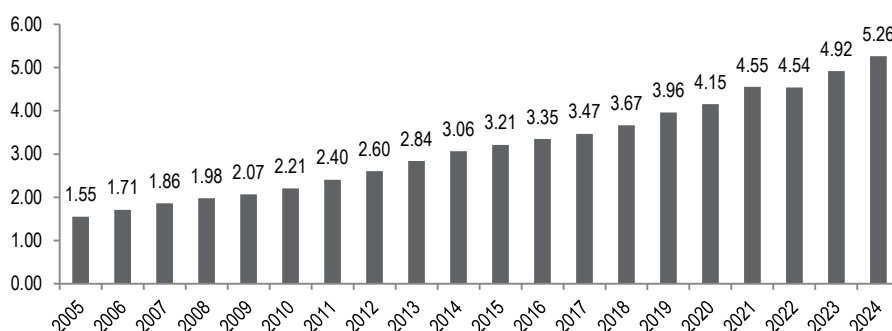
Figure 37: European Payment Penetration

V/M/A Volume % of GDP.



Note: GDP includes EU, emerging and developing Europe, Norway and the U.K.
Source: The IMF, company reports, J.P. Morgan estimates.

Figure 38: European Cards per Capita



Source: The World DataBank, company reports, J.P. Morgan estimates.

PSD2 and Open Banking

PSD2 (Revised Payment Services Directive) was implemented across the EU in early 2018. The goal of PSD2 was four-fold: (1) promote a more integrated and efficient European payments market, (2) level the playing field for payment service providers (including non-banks), (3) make payments more secure, and (4) reduce consumer costs. The directive includes more than 100 articles covering (1) customer authentication, (2) account access, (3) market places, and (4) surcharging.

Payment Service Directive Timeline

The EU's first Payment Service directive, which set rules regarding electronic transfers and payments in 30 European countries, went into effect in 2007 and was fully implemented in 2009, harmonizing pricing and improving payment security across the EU. In 2013, the European Commission revised the original PSD to take into account new types of payment services (such as Payment Initiation Services). PSD2 was adopted by the European parliament in November 2015 and went into force in January 2016. PSD2 went into force in early 2018. So far, the AISP and PISP markets have developed slowly, while banks have been less than eager to permit open API integrations.

PSD2 in Practice

The biggest change, in our view, is the rule requiring banks to offer third-party service providers access to their customers' accounts in order to manage and analyze their

finances and initiate payments, essentially ending the monopoly banks have historically had on account holder transaction data and payments services.

To that end, PSD2 introduced two new players to the financial landscape: AISPs (Account Information Service Providers) and PISPs (Payment Initiation Service Providers).

- AISPs have access to banking account and transaction information. AISPs could use transaction data to analyze spending behavior or aggregate account information across several bank accounts.
- PISPs create a software bridge between the customer and retailers to effect payment, without requiring the customer to reveal its payment card details (e.g., P2P and bill payment).

PSD2 gives third-party AISPs the ability to leverage customer spending and expense data. However, the Payment Initiation Service Providers (PISPs) stand to gain the most, in our view, as they will be able to enter the financial services market without the heavy compliance and infrastructure banks are required to maintain. These are early days, and we still don't know what these AISPs and PISPs will ultimately look like, but we've identified four broad outcomes:

1. Financial markets remain primarily domestic, and the use of non-banks for financial services will be limited.
2. Consumers adopt non-bank financial services, but markets remain primarily domestic.
3. Consumers adopt non-bank financial services across country lines.
4. Consumer trust in non-bank entities remains low; big banks become leading PISPs and AISPs.

Visa and Mastercard Reduce EU Cross-Border Interchange

In 2019, Visa and Mastercard agreed to reduce interchange fees on purchases made within the EU with foreign-issued debit and credit cards to 20bps and 30bps, respectively, while capping the rate for online (card not present) debit and credit transactions at 115bps and 150bps, respectively. All told, interchange rates were reduced by approximately 40%. The new rates went into effect in October 2019. Although Visa and Mastercard's issuer fees are not directly tied to interchange rates, this decision could ultimately pressure Visa and Mastercard's issuer fee economics.

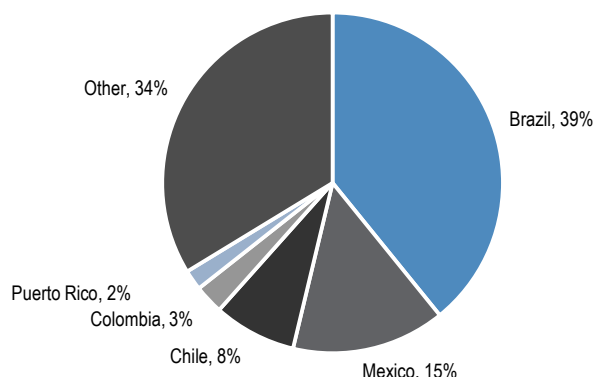
Dynamic Currency Conversion (DCC) allows a cardholder to pay for or withdraw money in their home currency while abroad. The service is typically offered by third-party operators (rather than the actual card issuer), which, in most cases, earn a fee and a currency conversion spread. Historically, fee disclosure was poor, making it hard for consumers to compare the cost of completing transactions in the local or home currency. The Commission recently amended rules to enhance transparency. While there were no formal caps on currency conversion rates, it will be interesting to see how increased disclosure impacts market pricing.

The Latin American Payments Market

Latin America generated roughly \$1.5 trillion in total purchase volume (measuring Visa, Mastercard, Amex and Diners) and almost 55 billion transactions in 2023. Of the networks, Visa was the dominant provider in Latin America with 58% market share, followed by Mastercard at 36%. By country, Brazil, Mexico and Chile are the largest markets, generating slightly more than 60% of bankcard purchase volume, extrapolating data from the region's top 50 issuers. By our estimates, bankcard payments represent 22% and 12% of Brazil and Mexico's GDP, respectively, versus 32% in the United States. As shown in the figure below, overall Latin American payments volume penetration was ~22% in 2024, up significantly from low-teens penetration of GDP pre-pandemic.

Figure 39: Latin America Purchase Volume by Country

Estimated share for 2023.

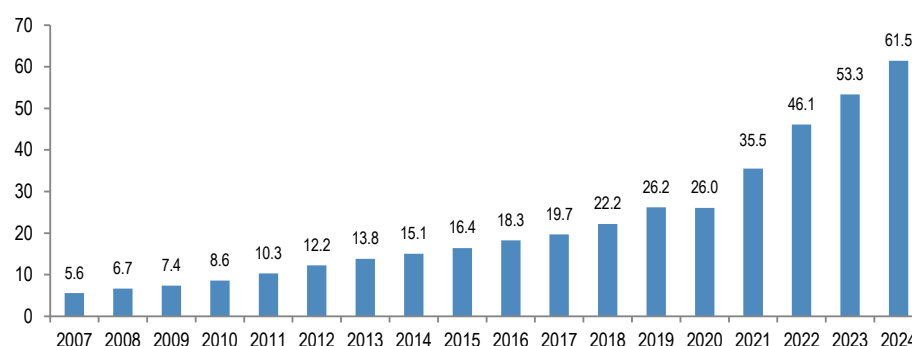


Note: Country-specific bankcard volume estimated based on data from the region's top-50 issuers.
Source: The Nilson Report, company reports, J.P. Morgan calculations.

The figures below show historical Latin American bankcard purchase volume and transaction trends. We note that bankcard purchase volume (in USD) and transactions have increased at 12% and 15% annualized rates, respectively, since 2007 with currency dynamics largely impacting the delta.

Figure 40: Latin America Bankcard Transactions

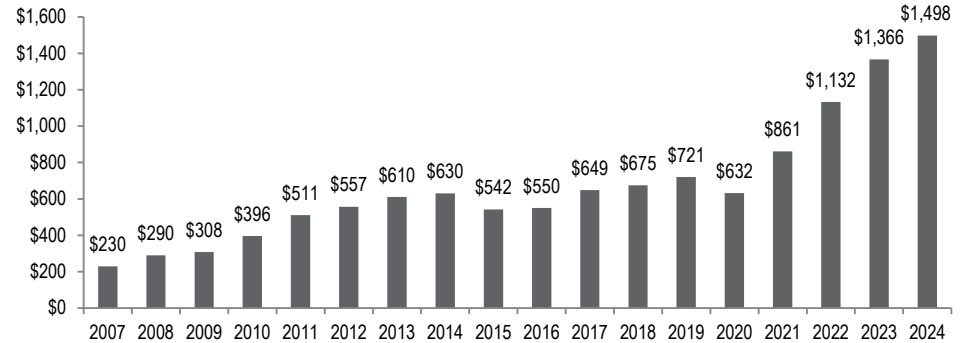
Transactions in billions.



Source: Visa and Mastercard company reports.

Figure 41: Latin America Bankcard Purchase Volume

\$ in USD billions.

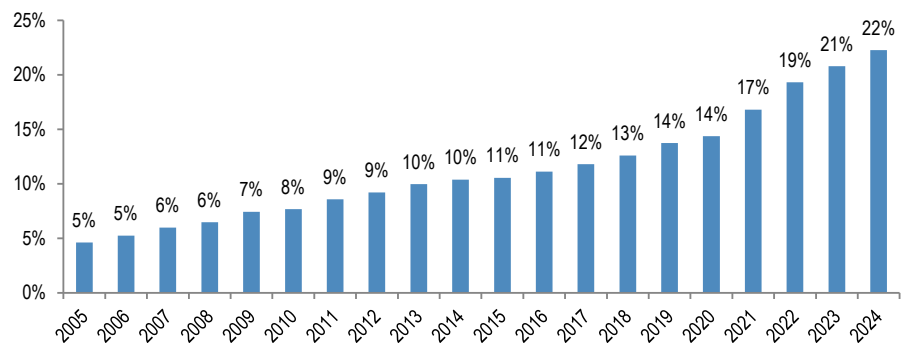


Source: Visa and Mastercard company reports.

The figures below show Latin American payment penetration and cards per capita over time. Payment penetration is calculated as V/MA purchase volume divided by GDP, while cards per capita is defined as total V/MA cards divided by the population over 15 years old. LatAm payment penetration has grown at a 9% annualized rate since 2005, the fastest of the regions examined and ahead of LatAm GDP growth. Cards per capita have increased at a 7% annualized rate since 2005, with a particular spike in 2021-2024 (13% CAGR since 2019).

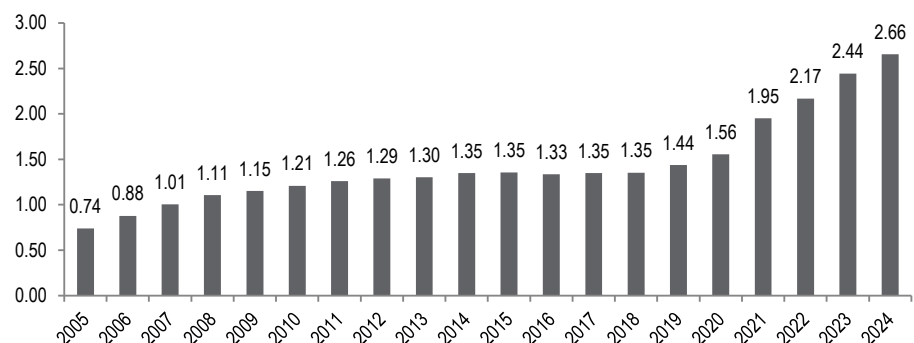
Figure 42: Latin America Payment Penetration

V/MA Volume % of GDP.



Source: The IMF, company reports, J.P. Morgan estimates.

Figure 43: Latin America Cards per Capita



Source: The World DataBank, company reports, J.P. Morgan estimates.

This section was contributed by:

Latin American Financials

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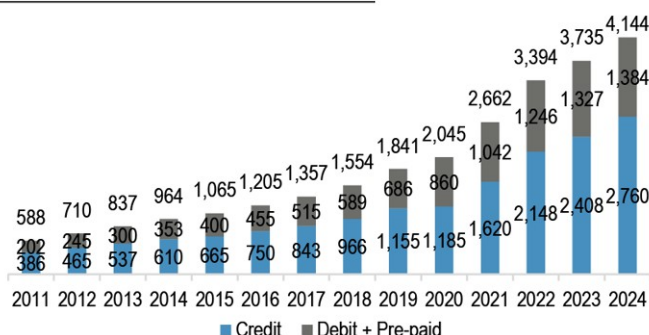
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A Closer Look at the Brazilian Card Market

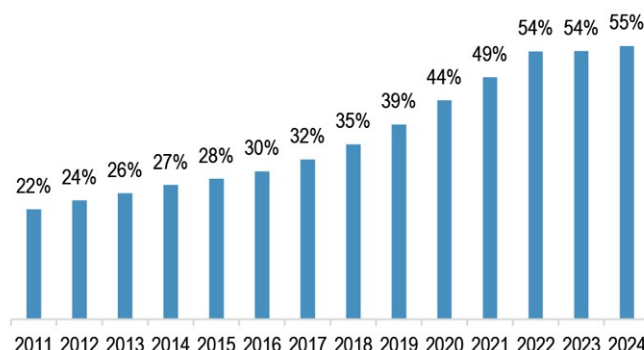
Brazilian market reached R\$4.1 trillion TPV in 2024 (~US\$758bn). This implies 11% Y-o-Y growth in 2024 (vs 10% in 2023), vs 10-year historical CAGR of ~16%. Growth was mostly supported by higher credit appetite from banks and secular trend of penetration from digital payments into PCE. On the other hand, PIX had been cannibalizing part of the volume growth potential, in our view.

In terms of penetration, card volumes now account for ~55% of Brazil's PCE in 2024 (personal consumption expenditures), up 1 p.p. vs 2023. The 16% Y-o-Y TPV 10-year CAGR implies historically a ~200bps increase in card penetration per year. The chart below does not include PIX P2B (R\$2.6 trillion), which would drive the penetration over 100% of PCE – given we have limited visibility on which PIX transactions are payment related vs. a replacement of wire transfer we prefer to just explore credit and debit below.

Figure 45: Card Volume % of Private Consumption



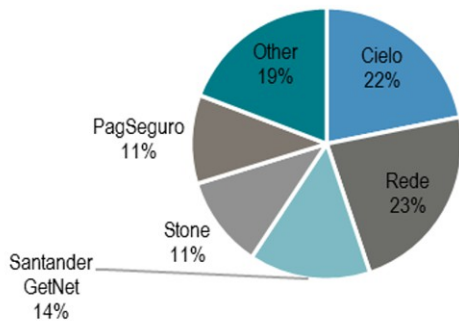
Source: ABECS.



Source: ABECS, IBGE.
Note: Does not include PIX P2B.

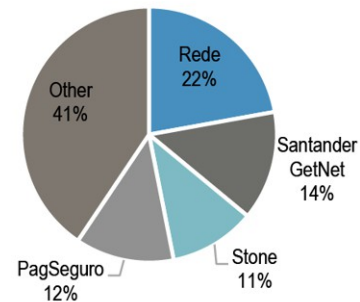
Below, we summarize market share for the largest Brazilian acquirers Rede, Cielo, Santander GetNet as well as newcomers Stone and PagSeguro. We believe Mercado Pago, from Meli (Neutral, covered by Marcelo Santos) ranks 6th in Brazil nowadays. We have seen a slowdown in market share gains from newcomers PAGS and STNE in recent years (i.e., PAGS +200bps and Stone flat in 2024).

Figure 46: TPV Market Share (2023)



Source: Company reports, ABECS, J.P. Morgan research.

Figure 47: TPV Market Share (2024)



Source: Company reports, ABECS, J.P. Morgan research.

Up until 2010 the former Redecard (currently called Rede) had the de facto monopoly of MasterCard transactions in Brazil, whilst Cielo (former VisaNet) had the contractual monopoly of Visa transactions. Those two companies had been controlled by the largest banks operating in Brazil: Redecard was created as a JV between Itau, Unibanco (Itau and Unibanco merged in 2008) and Citibank; and VisaNet was founded by Visa, Bradesco, Banco do Brasil, Banco Real (which later on became Santander Brasil) and Banco Nacional.

After the end of the exclusivities of Visa and MasterCard, several players made attempts to enter the Brazilian market. As already mentioned Santander Brasil was one of the first with Getnet. While Rede was initially a public traded company it was delisted in 2012 and today is fully controlled by Itau-Unibanco (OW, covered by Yuri Fernandes) while Cielo was delisted in 2024 and today has no minority, being fully controlled by Bradesco and Banco do Brasil (both N-rated, covered by Yuri Fernandes). Having acquirers inside banks has been shifting competitive dynamics. We note a gradual movement of sharing more economics with banking products (i.e., credit, cash management, deposits, etc.) which makes monetization only as an acquiring tougher. As a result we also see independent players becoming more and more banks / increasing banking tools.

Other players are:

1) **Stone**, a listed player with ~11% mkt share. Created by a group of investors that had expertise in the Brazilian card industry, it isn't linked to any retail bank. Today it is mostly a corporation. Its primary focus was e-commerce, but it has been growing fast in bricks-and-mortar merchants. After the acquisition of Elavon, its size more than doubled.

2) **PagSeguro**, or **PagBank** a listed player with ~12.5% mkt share. Owned by UOL media channel and has been gaining ground by selling POS, rather than renting them, and focusing on the low-end of the merchant pyramid (micro-merchants and autonomous professionals). In 2019 PAGs launched its digital bank, PagBank, which aims at expanding in the long tail of underserved retail clients.

3) **MercadoPago**, estimated at ~10% market share. Controlled by MercadoLibre, the biggest online retailer in Latin America. Since early 2018, we have seen the holding

company putting more focus on its payments solution arm, mostly focused on e-commerce and long-tail market.

4) **Banrisul/Vero**, a listed mid-sized bank controlled by the state of Rio Grande do Sul with ~1% mkt. share. The bank already had its own card brand, called Banricompras, of which it was the sole acquirer. After the market opening in 2010, it started competing more directly with Cielo and Rede.

5) **SafraPay, First Data, Global Payments, iZettle, SumUp** and other players, which we estimate have a very small portion of the market.

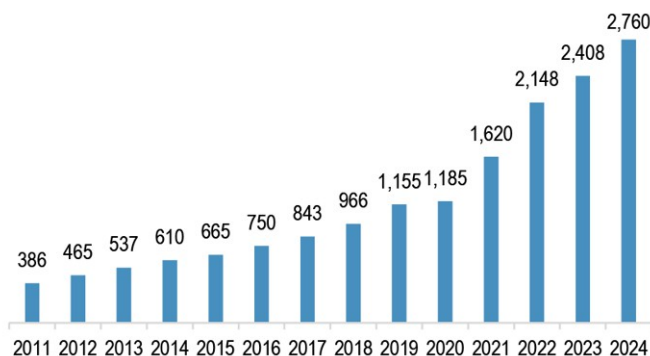
We see similar disruption on the issuing side, as new players build their business models around the profitable card industry in Brazil. Among the successful players, we highlight Nubank, PagSeguro, Inter, C6 Bank, and others. Particularly, market share on the issuing side (based on credit TPV) for top 4 incumbent banks declined from 87% in 2017 to ~59% in 2024.

Credit

Credit transactions account for ~67% of Brazil industry's TPV, or R\$2.8tn in 2024. Historically, credit volume growth averaged ~17% yearly growth in the last 10 years. In 2024, credit TPV grew ~15% vs 12% Y-o-Y in 2023 and 33% in 2022. In our view, the strong growth is the result of higher inflation, higher banks' credit appetite in the period and secular penetration of digital payments into PCE.

Figure 48: Credit TPV

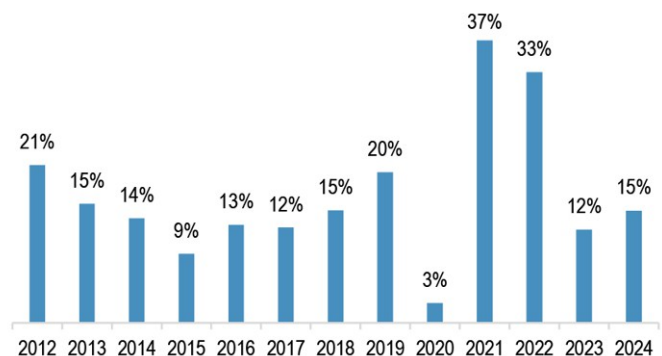
R\$ billions.



Source: ABECS.

Figure 49: Credit TPV

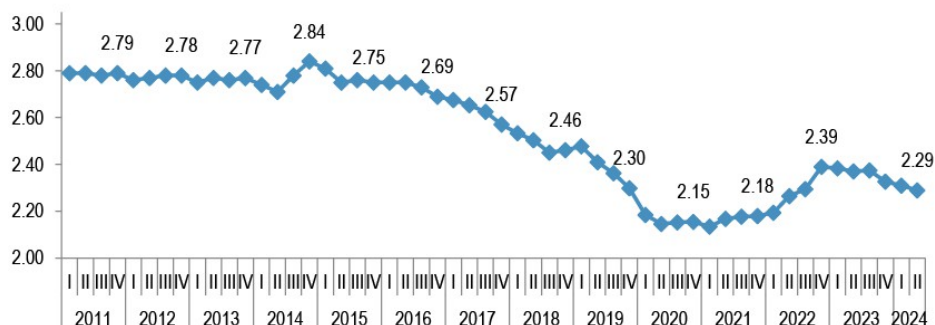
Y/Y growth.



Source: ABECS.

On pricing, Brazil saw significant compression on the industry's economics for merchant acquirers in the last years. Since the end of the monopoly in Brazil and increasing competition, gross merchant discount rates for credit declined from 2.95% in 2009 to ~2.3% in 24YE. MDRs net of interchange costs absorbed most of the hit, declining from 1.40% in 2009 to 0.58% in 24YE.

Figure 50: Gross Credit MDR (%)



Source: Brazil Central Bank.

Figure 51: Credit MDR Net of Interchange (%)



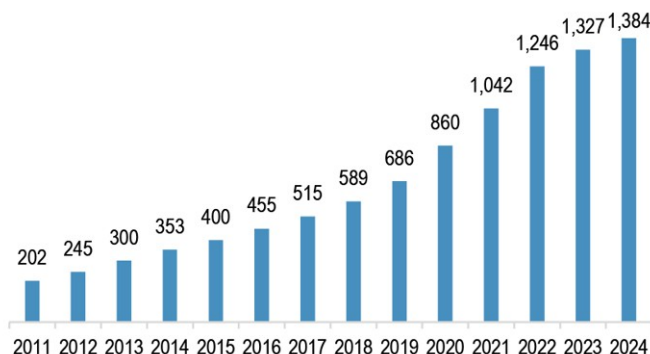
Source: Brazil Central Bank.

Debit

Debit transactions (including prepaid) account for ~33% of Brazil industry's TPV, totaling R\$1.4tn in 2024. Historically, debit volume growth averaged ~16% in the last 10 years. After growing 20% Y-o-Y in 2022, debit TPV growth decelerated to 6% in 2023 and 4% in 2024. We believe part of the deceleration is explained by cannibalization of PIX in Brazil, which should continue going forward. All in, we see PIX driving lower TPV growth vs historical average for debit going forward.

Figure 52: Debit + Pre-Paid TPV

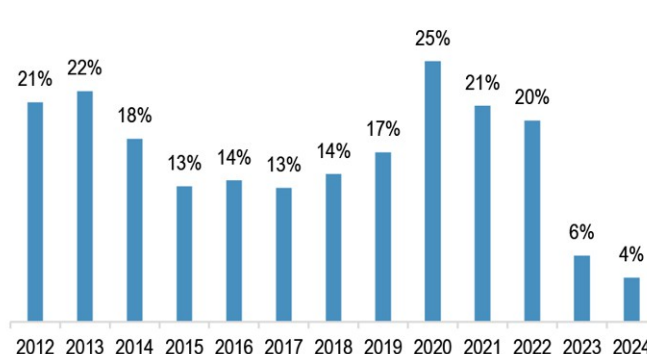
R\$ in billions.



Note: Excludes COVID-19 government stimulus checks.
Source: ABECS.

Figure 53: Debit + Pre-Paid

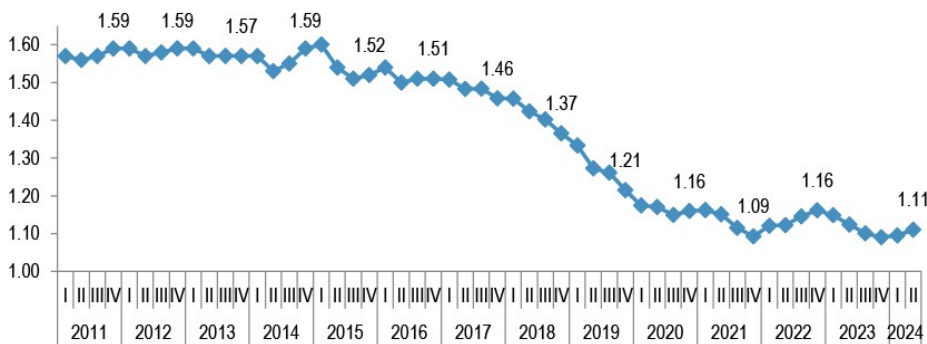
Y/Y growth.



Note: Excludes COVID-19 government stimulus checks.
Source: ABECS.

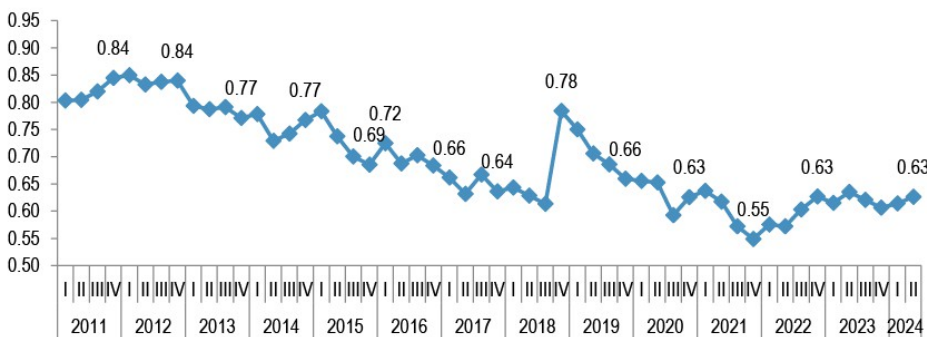
On pricing, we note significant compression in the industry’s economics for debit as well in recent years. While debit merchant discount rates were more stable from 2008-2014 around ~1.60%, it has significantly declined to ~1.10% in the last 10 years. Importantly, we note Brazil Central Bank introduced a cap of 0.50% (average) for debit interchange in Brazil since Oct-2018. This drove net MDRs to spike right after the regulation kicked in, but the reduction in interchange costs was gradually passed through to prices by acquirers. A new interchange cap of 0.50% maximum per transaction (instead of average) was introduced in April 2023, together with a 0.70% cap in pre-paid cards.

Figure 54: Gross Debit MDR (%)



Source: Brazil Central Bank.

Figure 55: Debit MDR Net of Interchange (%)



Source: Brazil Central Bank.

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Key Discussion Themes in Brazil

In our view, regulatory noise eased in the last years as a consequence of already fierce competition seen on both merchant acquiring and issuing business. More recently, the main discussed themes include: (i) interchange cap for pre-paid cards (it was already implemented – we also sometimes hear discussions about credit card caps); (ii) capital requirements for payment institutions (some regulation enacted in past years); (iii) food vouchers transitioning into open-schemes (discussions ongoing). Projects already launched but also important to be tracked include (i) PIX, the Central Bank instant payments platform, which has implications on networks, bank fees, debit costs as well as potentially credit and prepayment in the future; (ii) the centralized platform for receivables registration, which is still experiencing technical difficulties.

Direct regulation on prices (interchange cap)

i. Debit. In Brazil, the Central Bank proposed in March 2018 ([link](#)) a cap on average interchange fees paid by acquirers on debit transactions. Specifically, the regulation limited average debit interchange fee to 50bps, and maximum to 80bps – vs industry average in 2017 of ~80bps. The regulation kicked in on October 1st, 2018. In September 2022 ([link](#)), Central Bank introduced a new cap of maximum 50bps for debit, instead of average. This change was implemented in April 2023.

ii. Pre-paid Credit – Debit-like. Post interchange cap was enacted on debit cards, several issuers have migrated to prepaid credit cards. Those cards have debit-like characteristics – i.e., lack of credit risk, instantaneous withdrawal of transacted notional from cardholder's account – but enjoyed ~1.2% interchange (higher than the 0.8% debit cards had pre-regulation). In September 2022 ([link](#)), Central Bank proposed a new cap of 70bps for pre-pays cards, which started being implemented in April 2023.

iii. Credit. In 2018, former Central Bank governor mentioned it was studying a potential cap on credit interchange ([link](#)) but this never went through, and the noise meaningfully reduced in recent years. However, more recently a new deputy governor of Central Bank revisited this topic, mentioning some changes could eventually happen ([link](#)). In addition to that, more recently we saw regulation just out from Chile ([link](#)) as well as some noise that Central Bank could study the matter again. For more details, please refer to our piece [“Brazilian Financials: Interchange Cap on Credit – Impacts, Winners and Losers.”](#)

New Features Inside Instant payment platform (PIX)

On April 2018, the Central Bank announced it was studying the creation of a centralized instant payments platform and on November 2020 it became live. Particularly, the new platform called PIX operates 24/7, and is ultimately managed by the Central Bank. The platform targeted both person-to-person transactions as well as person-to-merchant transactions. The cost of transaction is less than R\$0.01 in the system.

We saw initially higher adoption of PIX on P2P transactions, which proved to be a threat to banks' current account fees and payment charges. Adoption in P2M has also gradually gained traction, hurting network fees as well as debit interchange fees.

In 2025, we see implementation of several different new features, including: (i) PIX offline and acceptance of PIX on POS machines through NFC; (ii) PIX Guaranteed, which should allow payments in installments; (iii) PIX receivables as guarantee for collateral. We also heard select industry participants trying to build cards with PIX rails

inside of them. It is still unclear how those features will fit the platform in the future (and if they will fit at all), and depending on how this unfolds, it could have broader implications to the credit value chain and more importantly, to the prepayment business. We discuss some of the implications of a PIX credit model in our note: [“Brazilian Financials: PIX Credit - Comparing Different Models and Assessing Risks.”](#)

Centralized registration for receivables

In December 2016 the Central Bank announced its intention to create a centralized registration platform for receivables (related to credit cards and others). According to the regulator, that would lead to lower cost of credit for companies as their collateral would be deemed safer by banks.

In late 2017 all credit card transactions started being registered at CIP (Camara Interbancaria de Pagamentos), a financial market settlement infrastructure controlled by Brazilian banks (SCG – Guarantee Control System). Nevertheless, several newcomers and later Cielo decided to leave the system as they were handicapped by an agreement in which banks could lock receivables of a merchant rendering them impossible to be prepaid.

The move drove the Central Bank to put forth a new resolution, in which every acquirer should be connected to a centralized clearing, although not necessarily the old system SCG. Today, approved registration companies are: CERC, CIP, TAG, CRDC, B3. The deadline for the implementation of its final structure was in June 2021, but the system faced several implementation issues (including interoperability issues) and is not fully operational. The Central Bank improved the regulation, publishing a new measure in December 2022 establishing new parameters for receivables registration and negotiation as well as pricing.

Under the final model, there was a fear that market participants would have easier access to merchants’ future receivables, making spreads on those flows lower and therefore hampering profitability on prepayment. However, we did not see this risk materializing in our view due to: (i) high cost of consultation of receivables agenda; (ii) higher penetration of automatically prepaid receivables (which are “born” as already ceded). Recently, Central Bank put out a public hearing willing to reduce consultation costs of those registration chambers.

Capital Requirements for Payment Institutions (IPs) and Prepayment RWA

Amidst claims of regulatory asymmetry by incumbent banks, the Central Bank launched a public hearing in November 2020 suggesting new capital requirements for financial institutions licensed as “Payment Institutions.” Particularly, minimum capital requirements today are based on 2% of monthly TPV (average for the last 12 months). In March 2022, the Central Bank published the final resolution establishing the new requirements and segregating financial entities in three clusters: (i) Type 1 (Financial conglomerates with payment subsidiaries); (ii) Type 2 (Pure payment operators); (iii) Type 3 (Payment institutions with other financial subsidiaries). The new rules establish minimum capital rules based on RWA/basel and proportional to size. The implementation was phased out until 2025.

For 2025, we also have discussions related to increasing the RWA – specifically operational risk factor – for prepayment operations. This could have a negative impact on prepayment capital requirements as we discussed in [“Brazilian Payments –](#)

Estimating Capital and Long-Term Potential.”

Payments in installments (“Crediario” product) – Buy Now Pay Later

In early 2018, ABECS (card industry association, whose members are mainly incumbents) proposed together with the regulator an alternative to payments in installments ([link](#)). More specifically, ABECS proposed a standardized product of installments with interest to be offered to merchants with a settlement in two to five days. We also started to see fintechs trying to offer similar BNPL products.

We believe that this type of initiative is unlikely to succeed ([link](#)) as i) 5-6% of GDP is currently driven by 0% APR product, and ii) other alternate products of interest-bearing installments, or “parcelado com juros,” already exist and are relatively unsuccessful. Ultimately, we already view the current 0% APR product as an economic response to the abnormally high consumer rates in Brazil. Adoption to the BNPL products so far has been low and impact on players and system small.

Food vouchers

Several unions in Brazil have agreements in which the employer grants food or gas vouchers to its employees. These are prepaid cards that can be used only in particular kinds of merchants (like restaurants and gas stations).

The potential issue is that voucher companies, such as Ticket and Alelo, compete with each other for contracts with large employers. On the other hand, merchants need to accept those vouchers; otherwise they would lose sales to competitors that accept them.

Such unevenness of bargaining power (skewed to employers / large companies over merchants) generates in some cases a cross-subsidy: employers getting discounts (e.g., paying the voucher company R\$99 for a R\$100 charge in the prepaid card), and merchants having to pay MDRs higher than in regular credit card transactions. Indeed, voucher MDRs can go north of 500bps vs. the average of no more than 300bps for credit cards. Voucher companies used to be closed-end schemes and acquirers operate with them in the VAN model.

However, new legislation from 2021/2022 moved food vouchers toward an open-scheme and prohibited the rebate structure to corporates (~2-5% on average of contract value). It also adds the possibility of portability. Most of the new rules are live and we expect the effective implementation of new systems in the short term.

The “D + 2” issue

Settlement period is very particular in Brazil. In the past, the lag between credit card transactions and the actual flow of funds to merchants in 30 days (the so-called D + 30) was seen as an anomaly by the Brazilian Central Bank, which states that a proprietary analysis indicates shorter floats would be advantageous to the market ([link](#)). The discussion brought concerns to industry players as either banks or acquirers would cope with working capital costs of paying the acquirers in advance (D+2 while receiving it in D+30). The regulator indicated at the time it was studying the change, and that it would like the transition to shorter settlement periods to be gradual in order to minimize side effects.

We saw, however, some of the players proactively announcing the D+2 settlement without additional costs to merchants (Rede – [link](#) and GetNet – [link](#)) in an attempt to

fence off competition. Between Rede and GetNet we saw players representing ~50% of the market moving into the D+2 direction. Other players offered D+2 but at the cost of some additional MDR.

Given the vast majority of the industry started to offer a D+0 / D+2 settlement option, the Central Bank reduced its willingness to intervene towards a shorter settlement period. This noise meaningfully reduced in the last years, and we don't foresee any changes on this item.

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